SAMPLE: SYNTHETIC DATA Mini-grid site performance GI-MSP-2026-09-F89BE511

GRIDINTEL · MINI-GRID INTELLIGENCE

Mini-grid site performance

The sites side by side, then each site: energy flow, PV, battery and generator, supply and interruptions, sales and collection.

SAMPLE: SYNTHETIC DATA

This report is a sample. It shows the format and the methods GridIntel would deliver; it is not a report of real results. Every figure in it is calculated from a designed demonstration dataset, and none describes a real network, site, customer or organization.

A designed demonstration dataset for September 2026. It describes no real network, customer, meter reading, outage, bill or payment.

FOR
Mini-grid operator

RECORDS OF
Demonstration Mini-grid Operator (synthetic)

PERIOD
September 2026

REPORT ID
GI-MSP-2026-09-F89BE511

FIGURES COMPUTED
20 Oct 2026, 10:10 WAT, from data as of 1 Oct 2026, 00:00 WAT

DEFINITION
Version 1

DATA FINGERPRINT (SHA-256)
cbb04117b9212f805ca1e47883b262174e4c8ceacfabc3337ad1f233cbdfa1a6

IN BRIEF

SITES IN THE REGISTER
4 • C
OK · Calculated

AVAILABILITY, DRY RIVER
95.8% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

AVAILABILITY, GRANITE HILL
99.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

AVAILABILITY, PALM LANDING
98.8% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

AVAILABILITY, TAMARIND CROSS
99.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Every recorded interruption counted, planned or not.

RENEWABLE FRACTION, DRY RIVER
49.6% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

RENEWABLE FRACTION, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

RENEWABLE FRACTION, PALM LANDING
62.3% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

RENEWABLE FRACTION, TAMARIND CROSS
94.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

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1 The sites side by side

How each site's plant, supply and sales compare.

Plant and energy, by site

	CONNECTIONS	RENEWABLE FRACTION	PERFORMANCE RATIO	PV CURTAILED	FUEL PER KWH	AVAILABILITY
Dry River 45 kWp PV · 90 kWh battery · 60 kVA diesel generator	300 • C	49.6% • C	75.2% • C	1.0% ≈ C	0.51 L/kWh • C	95.8% • C
Granite Hill 60 kWp PV · 150 kWh battery · no generator	130 • C	100.0% • C	21.8% • C	72.4% ≈ C	No generator	99.7% • C
Palm Landing 80 kWp PV · 200 kWh battery · 40 kVA diesel generator	393 • C	62.3% • C	76.5% • C	0.6% ≈ C	0.35 L/kWh • C	98.8% • C
Tamarind Cross 100 kWp PV · 240 kWh battery · 50 kVA diesel generator	465 • C	94.7% • C	74.4% • C	5.8% ≈ C	0.35 L/kWh • C	99.7% • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

Sales and collection, by site

Energy sold is energy billed. Cash basis for collection.

	ENERGY SOLD PER CONNECTION	COLLECTION EFFICIENCY
Dry River	25.9 kWh • C	97.3% • C
Granite Hill	11.6 kWh • C	93.4% • C
Palm Landing	19.7 kWh • C	88.3% • C
Tamarind Cross	22.5 kWh • C	96.0% • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0); GridIntel reference site performance (gridintel.site.reference 0.1.0).

Report differs from the records: Dry River. Dry River's report differs from its records on 3 of the 8 figures it states. Rule: A reported figure on the same basis as the calculation, and more than 1% of its own value away from it. The largest difference is shown first, measured against the reported figure itself, so that figures in different units can be set side by side.

Report differs from the records: Palm Landing. Palm Landing's report differs from its records on 2 of the 8 figures it states. Rule: A reported figure on the same basis as the calculation, and more than 1% of its own value away from it. The largest difference is shown first, measured against the reported figure itself, so that figures in different units can be set side by side.

Most fuel per kWh: Dry River. Dry River's generator burns the most fuel for each kWh it makes, and ran the longest. Rule: Of the sites with a generator, the one with the highest fuel per kWh, when it is at least 20% above the lowest.

PV held back for want of demand: Granite Hill. Granite Hill took less than half of the PV energy available to it. Its array is not underperforming: it is larger than its demand. Rule: At least 50% of the PV energy available to the site was curtailed.

Most revenue left uncollected: Palm Landing. Palm Landing left the most revenue uncollected. All of it is owed by 13 postpaid accounts; its prepaid accounts cannot owe. Rule: The site whose customer classes left the largest sum uncollected in the period, with the classes that owe it.

Who pays least. Across the portfolio, public institutions paid the smallest share of what they were billed. Rule: The customer class with the lowest collected-over-billed across every site, among classes with a shortfall.

2 Dry River

Dry River's energy, plant, supply and money; commissioned 2019-11-01.

Energy

RENEWABLE FRACTION, DRY RIVER
49.6% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, DRY RIVER
insufficient data !
Insufficient data · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, DRY RIVER
insufficient data !
Insufficient data · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Dry River

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	4,429 kWh • C
Generator generation	Adds	4,503 kWh • C
Energy generated	Equals	8,932 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	170 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	214 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	8,547 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	*insufficient data* !
Energy delivered What customer meters recorded.	Equals	*insufficient data* !

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

294 of 300 connections have a meter that recorded the whole period; 0 are unmetered and 6 incomplete.

Solar PV

PERFORMANCE RATIO, DRY RIVER
75.2% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, DRY RIVER
76.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, DRY RIVER
1.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, DRY RIVER
130.8 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, DRY RIVER
98.4 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, DRY RIVER
46 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (90 kWh, 40 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, DRY RIVER
18.1 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, DRY RIVER
90.1% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, DRY RIVER
177 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, DRY RIVER
39.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, DRY RIVER
20.1% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, DRY RIVER
1,795 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, DRY RIVER
1,625 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator (60 kVA, 48 kW, diesel)

FUEL PER KWH, DRY RIVER
0.51 L/kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Litres of fuel for each kWh the generator made.

RUNNING HOURS, DRY RIVER
315.0 h • C
OK · Calculated · gridintel.site.reference 0.1.0
The advance of the hour meter over the period.

FUEL USED, DRY RIVER
2,290 L • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR GENERATION, DRY RIVER
4,503 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR'S SHARE OF ENERGY, DRY RIVER
50.4% • C
OK · Calculated · gridintel.site.reference 0.1.0

Supply

AVAILABILITY, DRY RIVER
95.8% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, DRY RIVER
23.0 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, DRY RIVER
7.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, DRY RIVER
95.8% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Dry River

Every interruption recorded at the site in the period, in order.

	FROM	TO	HOURS	CAUSE	CUSTOMERS AFFECTED
Unplanned Fault at the powerhouse busbar.	6 Sep 2026, 14:00 WAT	6 Sep 2026, 17:00 WAT	3.0 h • C	Fault	300 • C
Unplanned Generator out of service and battery at its minimum: site disconnected.	18 Sep 2026, 04:00 WAT	18 Sep 2026, 09:00 WAT	5.0 h • C	Fault	300 • C
Unplanned Generator out of service and battery at its minimum: site disconnected.	18 Sep 2026, 10:00 WAT	18 Sep 2026, 11:00 WAT	1.0 h • C	Fault	300 • C

	FROM	TO	HOURS	CAUSE	CUSTOMERS AFFECTED
Unplanned Generator out of service and battery at its minimum: site disconnected.	18 Sep 2026, 15:00 WAT	18 Sep 2026, 16:00 WAT	1.0 h • C	Fault	300 • C
Unplanned Generator out of service and battery at its minimum: site disconnected.	18 Sep 2026, 17:00 WAT	18 Sep 2026, 23:00 WAT	6.0 h • C	Fault	300 • C
Unplanned Generator out of service and battery at its minimum: site disconnected.	19 Sep 2026, 00:00 WAT	19 Sep 2026, 09:00 WAT	9.0 h • C	Fault	300 • C
Unplanned Generator out of service and battery at its minimum: site disconnected.	19 Sep 2026, 19:00 WAT	20 Sep 2026, 00:00 WAT	5.0 h • C	Fault	300 • C

Sales and collection

REVENUE BILLED, DRY RIVER
₦2,186,916 • C
OK · Calculated · gridintel.site.reference 0.1.0

REVENUE COLLECTED, DRY RIVER
₦2,128,526 • C
OK · Calculated · gridintel.site.reference 0.1.0
Cash basis: money received in the period.

ACTIVE CONNECTIONS, DRY RIVER
300 • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY SOLD, DRY RIVER
7,757 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy billed. For a prepaid account that is energy purchased, not energy used.

ENERGY SOLD PER CONNECTION, DRY RIVER
25.9 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

CONSUMPTION PER CONNECTION, DRY RIVER
insufficient data
Insufficient data · Calculated · gridintel.site.reference 0.1.0
From customer meters.

REVENUE PER CONNECTION, DRY RIVER
₦7,290 • C
OK · Calculated · gridintel.site.reference 0.1.0

PRODUCTIVE-USE SHARE OF ENERGY SOLD, DRY RIVER
40.2% • C
OK · Calculated · gridintel.site.reference 0.1.0

COLLECTION EFFICIENCY, DRY RIVER
97.3% • C
OK · Calculated · gridintel.atcc.reference 0.1.0
Cash basis: received in the period against billed in the period.

Tariffs. Tariffs are assumptions, not current published rates.

By customer class, Dry River

Cash basis: money received in the period against bills issued in the period.

	ACCOUNTS	BILLED	COLLECTED	COLLECTION EFFICIENCY
Public institutions 0 prepaid, 3 postpaid	3 • C	₦74,098 • C	₦15,708 • C	21.2% • C
Households 260 prepaid, 0 postpaid	260 • C	₦823,951 • C	₦823,951 • C	100.0% • C
Productive use (mills, workshops) 9 prepaid, 0 postpaid	9 • C	₦810,841 • C	₦810,841 • C	100.0% • C
Shops and businesses 28 prepaid, 0 postpaid	28 • C	₦478,026 • C	₦478,026 • C	100.0% • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0).

3 Granite Hill

Granite Hill's energy, plant, supply and money; commissioned 2024-09-01.

Energy

RENEWABLE FRACTION, GRANITE HILL
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, GRANITE HILL
20.7% • D
OK · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, GRANITE HILL
5.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Granite Hill

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	1,917 kWh • C
Generator generation	Adds	0 kWh • C
Energy generated	Equals	1,917 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	137 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	180 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	1,599 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	80 kWh • D
Energy delivered What customer meters recorded.	Equals	1,519 kWh • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

130 of 130 connections have a meter that recorded the whole period; 0 are unmetered and 0 incomplete.

Solar PV

PERFORMANCE RATIO, GRANITE HILL
21.8% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, GRANITE HILL
79.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, GRANITE HILL
72.4% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, GRANITE HILL
146.7 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, GRANITE HILL
31.9 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, GRANITE HILL
5,039 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (150 kWh, 40 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, GRANITE HILL
4.8 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, GRANITE HILL
90.7% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, GRANITE HILL
79 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, GRANITE HILL
92.5% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, GRANITE HILL
44.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, GRANITE HILL
852 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, GRANITE HILL
715 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

> **Generator.** Granite Hill has no generator in the register.

Supply

AVAILABILITY, GRANITE HILL
99.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, GRANITE HILL
23.9 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, GRANITE HILL
1.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, GRANITE HILL
99.7% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Granite Hill

Every interruption recorded at the site in the period, in order.

	FROM	TO	HOURS	CAUSE	CUSTOMERS AFFECTED
Unplanned Inverter trip.	16 Sep 2026, 08:00 WAT	16 Sep 2026, 10:00 WAT	2.0 h • C	Fault	130 • C

Sales and collection

REVENUE BILLED, GRANITE HILL
₦392,505 • C
OK · Calculated · gridintel.site.reference 0.1.0

REVENUE COLLECTED, GRANITE HILL
₦366,710 • C
OK · Calculated · gridintel.site.reference 0.1.0
Cash basis: money received in the period.

ACTIVE CONNECTIONS, GRANITE HILL
130 • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY SOLD, GRANITE HILL
1,507 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy billed. For a prepaid account that is energy purchased, not energy used.

ENERGY SOLD PER CONNECTION, GRANITE HILL
11.6 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

CONSUMPTION PER CONNECTION, GRANITE HILL
11.7 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
From customer meters.

REVENUE PER CONNECTION, GRANITE HILL
₦3,019 • C
OK · Calculated · gridintel.site.reference 0.1.0

PRODUCTIVE-USE SHARE OF ENERGY SOLD, GRANITE HILL
10.7% • C
OK · Calculated · gridintel.site.reference 0.1.0

COLLECTION EFFICIENCY, GRANITE HILL

93.4% • C

OK · Calculated · gridintel.atcc.reference 0.1.0
Cash basis: received in the period against billed in the period.

Tariffs. Tariffs are assumptions, not current published rates.

By customer class, Granite Hill

Cash basis: money received in the period against bills issued in the period.

	ACCOUNTS	BILLED	COLLECTED	COLLECTION EFFICIENCY
Public institutions 0 prepaid, 2 postpaid	2 • C	₦32,612 • C	₦6,817 • C	20.9% • C
Households 118 prepaid, 0 postpaid	118 • C	₦243,937 • C	₦243,937 • C	100.0% • C
Shops and businesses 9 prepaid, 0 postpaid	9 • C	₦78,935 • C	₦78,935 • C	100.0% • C
Productive use (mills, workshops) 1 prepaid, 0 postpaid	1 • C	₦37,021 • C	₦37,021 • C	100.0% • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0).

4 Palm Landing

Palm Landing's energy, plant, supply and money; commissioned 2021-06-01.

Energy

RENEWABLE FRACTION, PALM LANDING
62.3% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, PALM LANDING
18.7% • D
OK · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, PALM LANDING
14.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Palm Landing

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	5,946 kWh • C
Generator generation	Adds	3,600 kWh • C
Energy generated	Equals	9,546 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	302 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	216 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	9,028 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	1,264 kWh • D
Energy delivered What customer meters recorded.	Equals	7,764 kWh • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

417 of 417 connections have a meter that recorded the whole period; 0 are unmetered and 0 incomplete.

Solar PV

PERFORMANCE RATIO, PALM LANDING
76.5% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, PALM LANDING
77.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, PALM LANDING
0.6% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, PALM LANDING
97.1 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, PALM LANDING
74.3 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, PALM LANDING
37 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (200 kWh, 60 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, PALM LANDING
15.6 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, PALM LANDING
89.9% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, PALM LANDING
347 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, PALM LANDING
38.1% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, PALM LANDING
15.1% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, PALM LANDING
3,427 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, PALM LANDING
3,124 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator (40 kVA, 32 kW, diesel)

FUEL PER KWH, PALM LANDING
0.35 L/kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Litres of fuel for each kWh the generator made.

RUNNING HOURS, PALM LANDING
150.0 h • C
OK · Calculated · gridintel.site.reference 0.1.0
The advance of the hour meter over the period.

FUEL USED, PALM LANDING
1,248 L • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR GENERATION, PALM LANDING
3,600 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR'S SHARE OF ENERGY, PALM LANDING
37.7% • C
OK · Calculated · gridintel.site.reference 0.1.0

Supply

AVAILABILITY, PALM LANDING
98.8% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, PALM LANDING
23.7 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, PALM LANDING
2.0 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, PALM LANDING
99.2% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Palm Landing

Every interruption recorded at the site in the period, in order.

	FROM	TO	HOURS	CAUSE	CUSTOMERS AFFECTED
Unplanned Storm damage to the distribution network.	13 Sep 2026, 16:00 WAT	13 Sep 2026, 22:00 WAT	6.0 h • C	Weather	393 • C
Planned Planned line work.	25 Sep 2026, 09:00 WAT	25 Sep 2026, 12:00 WAT	3.0 h • C	Planned maintenance	393 • C

Sales and collection

REVENUE BILLED, PALM LANDING
₦2,076,678 • C
OK · Calculated · gridintel.site.reference 0.1.0

REVENUE COLLECTED, PALM LANDING
₦1,833,537 • C
OK · Calculated · gridintel.site.reference 0.1.0
Cash basis: money received in the period.

ACTIVE CONNECTIONS, PALM LANDING
393 • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY SOLD, PALM LANDING
7,758 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy billed. For a prepaid account that is energy purchased, not energy used.

ENERGY SOLD PER CONNECTION, PALM LANDING
19.7 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

CONSUMPTION PER CONNECTION, PALM LANDING
19.8 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
From customer meters.

REVENUE PER CONNECTION, PALM LANDING
₦5,284 • C
OK · Calculated · gridintel.site.reference 0.1.0

PRODUCTIVE-USE SHARE OF ENERGY SOLD, PALM LANDING
22.4% • C
OK · Calculated · gridintel.site.reference 0.1.0

COLLECTION EFFICIENCY, PALM LANDING
88.3% • C
OK · Calculated · gridintel.atcc.reference 0.1.0
Cash basis: received in the period against billed in the period.

Tariffs. Tariffs are assumptions, not current published rates.

By customer class, Palm Landing

Cash basis: money received in the period against bills issued in the period.

	ACCOUNTS	BILLED	COLLECTED	COLLECTION EFFICIENCY
Productive use (mills, workshops) 0 prepaid, 8 postpaid	8 • C	₦417,281 • C	₦270,141 • C	64.7% • C
Public institutions 0 prepaid, 5 postpaid	5 • C	₦121,254 • C	₦25,253 • C	20.8% • C
Households 364 prepaid, 0 postpaid	364 • C	₦1,041,117 • C	₦1,041,117 • C	100.0% • C
Shops and businesses 40 prepaid, 0 postpaid	40 • C	₦497,025 • C	₦497,025 • C	100.0% • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0).

5 Tamarind Cross

Tamarind Cross's energy, plant, supply and money; commissioned 2023-03-01.

Energy

RENEWABLE FRACTION, TAMARIND CROSS
94.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Share of energy generated.

SITE LOSS, TAMARIND CROSS
12.1% • D
OK · Derived · gridintel.site.reference 0.1.0
Energy generated that reached no customer meter, as a share of energy generated.

DISTRIBUTION LOSS, TAMARIND CROSS
7.0% • D
OK · Derived · gridintel.site.reference 0.1.0
Share of energy sent out.

Energy flow, Tamarind Cross

From generation to customer meters, top to bottom. Each line keeps its own origin: a residual is shown as one.

	IN THE CHAIN	ENERGY
PV generation	Adds	11,274 kWh • C
Generator generation	Adds	630 kWh • C
Energy generated	Equals	11,904 kWh • C
Net into storage Energy into the battery less energy out of it: storage loss and any change in what is stored.	Less	443 kWh • C
Station use A residual: the powerhouse's own use and anything unaccounted for.	Less	216 kWh • D
Energy sent out Metered where the powerhouse hands over to the network: the energy available to customers.	Equals	11,245 kWh • C
Distribution loss A residual: technical loss and unmetered use together.	Less	786 kWh • D
Energy delivered What customer meters recorded.	Equals	10,460 kWh • C

Method: GridIntel reference site performance (gridintel.site.reference 0.1.0).

465 of 465 connections have a meter that recorded the whole period; 0 are unmetered and 0 incomplete.

Solar PV

PERFORMANCE RATIO, TAMARIND CROSS
74.4% • C
OK · Calculated · gridintel.site.reference 0.1.0
Counts curtailment as a loss, as IEC 61724 does.

PERFORMANCE RATIO BEFORE CURTAILMENT, TAMARIND CROSS
79.0% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Adds the curtailment estimate back: what the array could do.

PV CURTAILED, TAMARIND CROSS
5.8% ≈ C
Estimated inputs · Calculated · gridintel.site.reference 0.1.0
Of the PV energy that was available.

IN-PLANE IRRADIATION, TAMARIND CROSS
151.5 kWh/m² • C
OK · Calculated · gridintel.site.reference 0.1.0
Measured on site; the input to the performance ratio.

SPECIFIC YIELD, TAMARIND CROSS
112.7 kWh/kWp • C
OK · Calculated · gridintel.site.reference 0.1.0

CURTAILMENT, TAMARIND CROSS
694 kWh ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0
The controller's estimate. Not subtracted from metered generation, which is already net of it.

Battery (240 kWh, 80 kW, lithium iron phosphate)

EQUIVALENT FULL CYCLES, TAMARIND CROSS
17.4 • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy out of the battery over its nameplate capacity.

ROUND-TRIP EFFICIENCY, TAMARIND CROSS
90.2% ≈ E
Estimated inputs · Estimated · gridintel.site.reference 0.1.0

STORAGE LOSS, TAMARIND CROSS
450 kWh ≈ D
Estimated inputs · Derived · gridintel.site.reference 0.1.0
The change in stored energy is estimated from state of charge and nameplate capacity.

STATE OF CHARGE, MEAN, TAMARIND CROSS
57.5% • C
OK · Calculated · gridintel.site.reference 0.1.0

STATE OF CHARGE, LOWEST, TAMARIND CROSS
15.9% • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY INTO STORAGE, TAMARIND CROSS
4,608 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY OUT OF STORAGE, TAMARIND CROSS
4,166 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

Generator (50 kVA, 40 kW, diesel)

FUEL PER KWH, TAMARIND CROSS
0.35 L/kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Litres of fuel for each kWh the generator made.

RUNNING HOURS, TAMARIND CROSS
21.0 h • C
OK · Calculated · gridintel.site.reference 0.1.0
The advance of the hour meter over the period.

FUEL USED, TAMARIND CROSS
218 L • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR GENERATION, TAMARIND CROSS
630 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

GENERATOR'S SHARE OF ENERGY, TAMARIND CROSS
5.3% • C
OK · Calculated · gridintel.site.reference 0.1.0

Supply

AVAILABILITY, TAMARIND CROSS
99.7% • C
OK · Calculated · gridintel.site.reference 0.1.0
Customer-weighted; every recorded interruption counted.

HOURS OF SUPPLY PER DAY, TAMARIND CROSS
23.9 h/day • C
OK · Calculated · gridintel.site.reference 0.1.0

INTERRUPTIONS PER CONNECTION, TAMARIND CROSS
1.3 • C
OK · Calculated · gridintel.site.reference 0.1.0

AVAILABILITY WITHOUT PLANNED INTERRUPTIONS, TAMARIND CROSS
100.0% • C
OK · Calculated · gridintel.site.reference 0.1.0

Interruptions, Tamarind Cross

Every interruption recorded at the site in the period, in order.

	FROM	TO	HOURS	CAUSE	CUSTOMERS AFFECTED
Planned Planned inverter service.	10 Sep 2026, 10:00 WAT	10 Sep 2026, 12:00 WAT	2.0 h • C	Planned maintenance	465 • C
Unplanned Fault on one low-voltage branch.	22 Sep 2026, 19:00 WAT	22 Sep 2026, 20:00 WAT	1.0 h • C	Fault	120 • C

Sales and collection

REVENUE BILLED, TAMARIND CROSS
₦2,672,889 • C
OK · Calculated · gridintel.site.reference 0.1.0

REVENUE COLLECTED, TAMARIND CROSS
₦2,566,180 • C
OK · Calculated · gridintel.site.reference 0.1.0
Cash basis: money received in the period.

ACTIVE CONNECTIONS, TAMARIND CROSS
465 • C
OK · Calculated · gridintel.site.reference 0.1.0

ENERGY SOLD, TAMARIND CROSS
10,450 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
Energy billed. For a prepaid account that is energy purchased, not energy used.

ENERGY SOLD PER CONNECTION, TAMARIND CROSS
22.5 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0

CONSUMPTION PER CONNECTION, TAMARIND CROSS
22.5 kWh • C
OK · Calculated · gridintel.site.reference 0.1.0
From customer meters.

REVENUE PER CONNECTION, TAMARIND CROSS
₦5,748 • C
OK · Calculated · gridintel.site.reference 0.1.0

PRODUCTIVE-USE SHARE OF ENERGY SOLD, TAMARIND CROSS
29.3% • C
OK · Calculated · gridintel.site.reference 0.1.0

COLLECTION EFFICIENCY, TAMARIND CROSS
96.0% • C
OK · Calculated · gridintel.atcc.reference 0.1.0
Cash basis: received in the period against billed in the period.

Tariffs. Tariffs are assumptions, not current published rates.

By customer class, Tamarind Cross

Cash basis: money received in the period against bills issued in the period.

	ACCOUNTS	BILLED	COLLECTED	COLLECTION EFFICIENCY
Public institutions 0 prepaid, 6 postpaid	6 • C	₦136,040 • C	₦29,331 • C	21.6% • C
Households 390 prepaid, 0 postpaid	390 • C	₦1,165,944 • C	₦1,165,944 • C	100.0% • C
Productive use (mills, workshops) 14 prepaid, 0 postpaid	14 • C	₦703,970 • C	₦703,970 • C	100.0% • C
Shops and businesses 55 prepaid, 0 postpaid	55 • C	₦666,935 • C	₦666,935 • C	100.0% • C

Method: GridIntel reference collection (gridintel.atcc.reference 0.1.0).

6 Method, sources and how to reproduce this report

Where every figure came from, and how a reader can check that this document is the one the records give.

Methodologies

METHODOLOGY	VERSION	WHAT IT SAYS OF ITSELF
GridIntel reference collection `gridintel.atcc.reference`	`0.1.0`	GridIntel reference calculation — not a regulatory figure.
GridIntel reference site performance `gridintel.site.reference`	`0.1.0`	GridIntel reference calculation — not a regulatory figure.

Sources

- Synthetic mini-grid vending and billing (synthetic)
- Synthetic mini-grid meter data (synthetic)
- Synthetic mini-grid plant monitoring (synthetic)
- Synthetic mini-grid site register (synthetic)
- Synthetic mini-grid site reports (synthetic)

> Every source above is synthetic: designed demonstration data, not records of a real organization.

How to reproduce it

1. Build the report with definition version 1, for September 2026, from the Demonstration Mini-grid Operator (synthetic) records.
2. The data fingerprint must be `cbb04117b9212f805ca1e47883b262174e4c8ceacfabc3337ad1f233cbdfa1a6`. SHA-256 over the sorted SHA-256 digests of every distinct result the report's services were given by the repositories, each written as JSON with its keys in order.
3. With the same definition, methodologies and fingerprint the report ID is the same: GI-MSP-2026-09-F89BE511. A different ID means something above differs.
4. The figures with their status, origin, unit and method are in the CSV of the same report ID: gridintel-site-performance-2026-09-SAMPLE-SYNTHETIC.csv.

7 Regulatory references

> How far this report carries the fields of the instruments below, as read from the instruments themselves. It is not a claim that the report conforms to any regulator's or programme's template: none has been checked against the template it would be filed on.

INSTRUMENT	REQUIREMENT	THIS REPORT
NERC Mini-Grid Regulations 2023 (NERC-R-117-2023), s. 13(2) and s. 21(2)	Reports in the form prescribed in Schedule 4, per mini-grid or combined for a portfolio	**NOT CARRIED** Schedule 4 is not in the published text read. This is the operator's own performance report, not a filing.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Renewable energy fraction	**CARRIED** Sections 1 and each site's section.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Percentage of time of available power supply at a site per month, as measured by inverter output	**PARTLY** Each site's section gives customer-weighted availability from interruption records, not from inverter output.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	Average consumption per user (ACPU); ARPU	**PARTLY** Each site's section gives consumption, energy sold and revenue per connection, for a month.
DARES Project Implementation Manual (REA, October 2024, rev. June 2025), Component 1, Table O.1	OPEX, EBITDA	**NOT CARRIED** Not held.

8 Glossary and key

Energy sent out	Energy metered where the powerhouse hands over to the network: what is available to customers.
Station use	A residual: the powerhouse's own use and anything unaccounted for.
Site loss	Energy generated that reached no customer meter, as a share of energy generated.
Curtailment	PV energy that was available but not taken, as the controller estimates it.
Equivalent full cycles	Energy out of the battery over its nameplate capacity.
Specific yield	PV energy per kWp of array rating.
Fuel per kWh	Litres of fuel for each kWh the generator made.
Availability	Customer-weighted share of the period with supply: 1 – (customer-hours interrupted ÷ customer-hours in the period). Every recorded interruption is counted.
Hours of supply a day	The average hours a day the average connection had supply in the period.
Connection	A customer connection in the site's register. An active connection is one billed in the period.
Energy sold	Energy billed. For a prepaid account it is energy purchased, which may be used later.
Consumption	Energy recorded by customer meters.
Renewable fraction	The share of energy generated that came from PV.
Performance ratio	PV energy delivered against what the array's rating and the measured irradiation would give; curtailment counts as a loss (IEC 61724).
Collection efficiency	Revenue collected in the period ÷ revenue billed in the period, on a cash basis.
Claim	A figure a site's own report states. It agrees when it is on the basis of the calculation and within the tolerance of it.
Tariff assumption	A rate used to price bills in this dataset that is not a published or approved tariff.
Measured	Read from a meter, a device or a source record.
Reported	A figure published by someone else, shown as they stated it. It is never altered.
Calculated	Computed by GridIntel from records, under a named and versioned methodology.
Estimated	An estimate, not a measurement; the method is named with it.
Derived	A residual: what is left when other figures are subtracted. It carries their uncertainty.
Insufficient data	An input the figure needs is missing. No value is given rather than a guess, and never a zero.
Not available	No source holds this figure, or the data does not cover the period.
Same basis	A reported figure and a calculation count the same things over the same period by the same rule. Only figures on the same basis are compared.
Data fingerprint	A SHA-256 digest of every record the report was built from. The same records give the same fingerprint; any change to them gives another.
Report ID	The report's code, its period and the first characters of a digest of its definition, methodologies and data fingerprint. The same report of the same data has the same ID.

The marks beside a figure

● **OK**	Every input is present and measured.
≈ **Estimated inputs**	Calculated; every input is present, but at least one was estimated rather than measured.
! **Insufficient data**	An input this figure needs is missing. No value is shown rather than a guess.
∅ **Not computable**	The inputs exist but the calculation is undefined for them.
– **Not available**	The platform has no source for this figure.
M **Measured**	Read from a meter, device or source record.
R **Reported**	Published by someone else; shown as stated.
C **Calculated**	Computed by GridIntel under a named methodology.
E **Estimated**	An estimate, not a measurement.
D **Derived**	A residual: what is left after subtracting other figures. It inherits their uncertainty.

SAMPLE: SYNTHETIC DATA. End of report GI-MSP-2026-09-F89BE511.